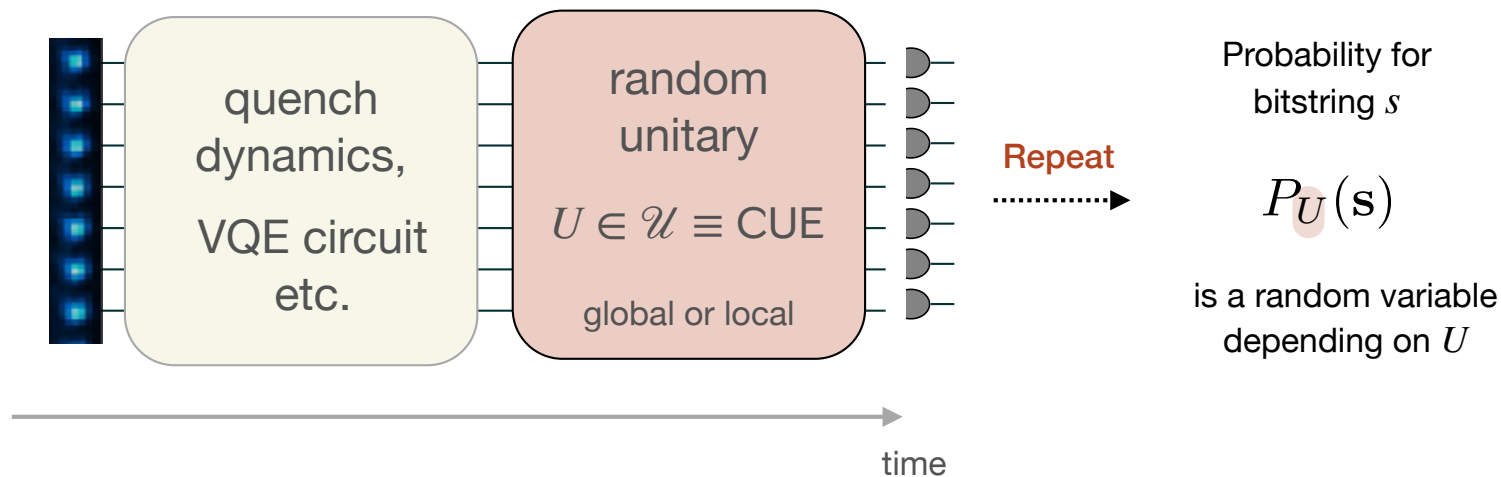
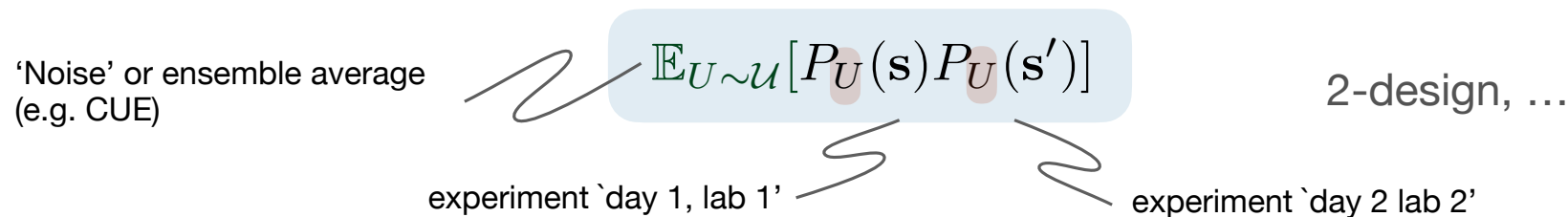


Randomized Measurements

Measurement post-processing



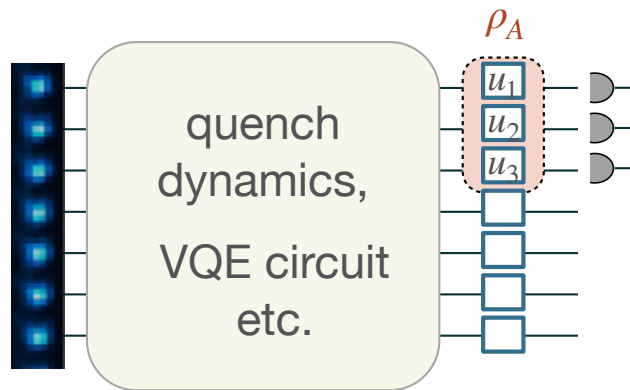
(Cross-) Correlation of probabilities



... hybrid classical-quantum protocols

Randomized Measurements

Measurement post-processing



(Cross-) Correlation of probabilities

$$\mathbb{E}_{U \sim \mathcal{U}}[P_U(s)P_U(s')]$$

experiment `day 1, lab 1`

experiment `day 2 lab 2`

- OTOCS

theory - B. Vermersch et al., Phys. Rev. X **9**, 021061 (2019).

exp - M. K. Joshi et al., Phys. Rev. Lett. **124**, 240505 (2020).

- topological invariants

theory - A. Elben et al., Science Advances **6**, eaaz3666 (2020).

- Partially transposed density matrix

theory [+ exp] - A. Elben et al., Phys. Rev. Lett. **125**, 200501 (2020).

theory [+ exp] - A. Neven et al., Npj Quantum Inf. **7**, (2021).

- Entanglement Hamiltonian Tomography

theory - C Kokail et al., Nat. Phys. **17**, 936 (2021).

theory + exp - MK Joshi, C Kokail, R van Bijnen et al., arXiv 2023

- Spectral form factor & quantum chaos

theory - L. K. Joshi et al., Phys. Rev. X. **12**, (2022).

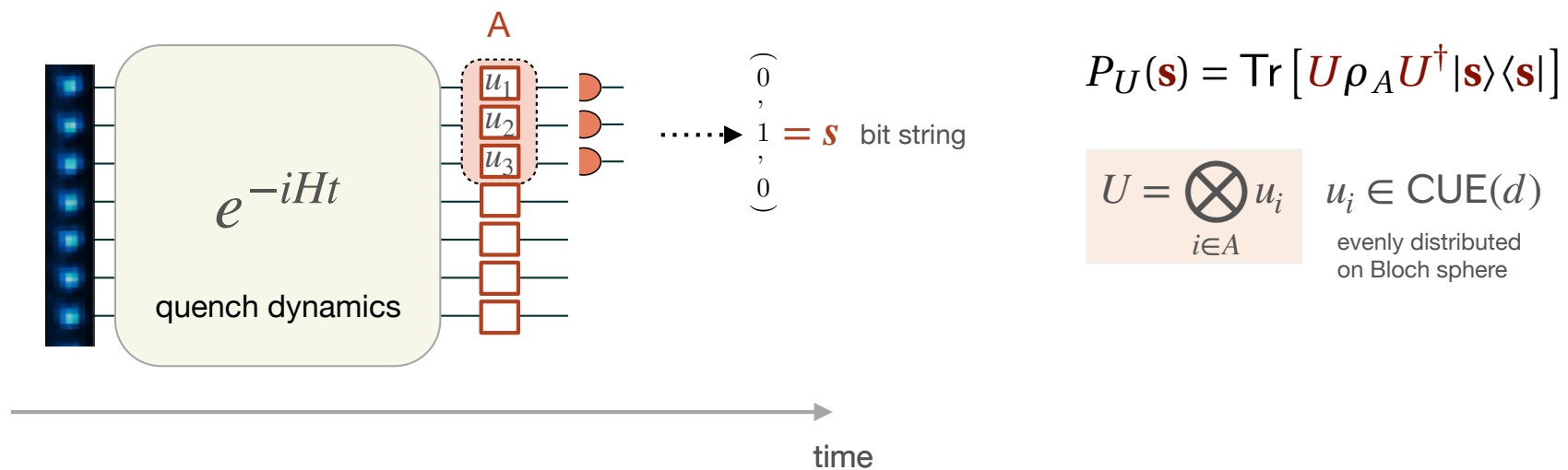
exp - L. K. Joshi et al. with Monroe group, unpublished

- observation of quantum Mpemba effect

theory + exp - MK Joshi et al. unpublished

Randomized Measurements: Local Random Unitaries

Measurement post-processing



purity - Renyi entropy

$$\text{Tr} \rho_A^2 = \mathbb{E}_{U \sim \text{CUE}}[\hat{P}_2] \quad \text{with} \quad \hat{P}_2 = 2^{|A|} \sum_{s, s'} (-2)^{-D[s, s']} \underbrace{P_U(\mathbf{s}) P_U(\mathbf{s}')}_{\text{cross-correlation}}$$

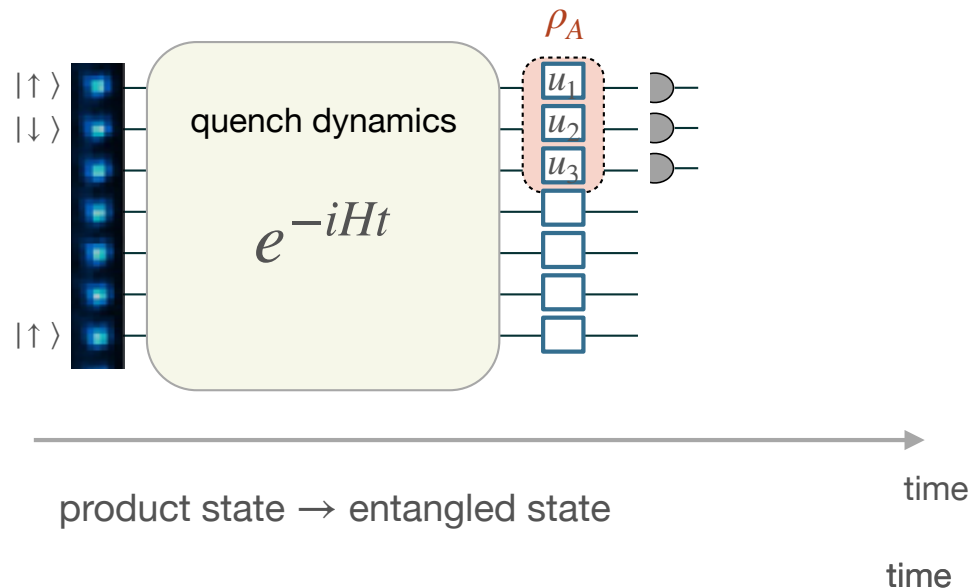
↑
Hamming distance

Random *single spin* rotations are sufficient!

T Brydges, A Elben et al., Science (2019)
A Elben, B Vermersch, et al., PRA (2019)

Randomized Measurements: Tomography

Quench dynamics with analog quantum simulator



Goal: learn operator structure of Entanglement Hamiltonian

$$\rho_A = e^{-\tilde{H}_A}$$

... learn efficiently only if we know something about EH

Randomized Tomography

$$\rho_A = \mathbb{E}_{U \sim \text{CUE}}[\hat{\rho}_A]$$

exponentially expensive



$$\hat{\rho}_A = \sum_{s,s'} \sum_U P_U(s) (-2)^{-D[s,s']} U |s'\rangle \langle s'| U^\dagger$$

tomographically complete



C Kokail



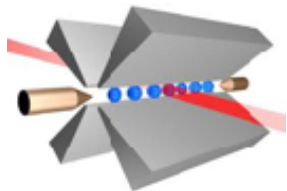
R van Bijnen



M Joshi

Exploring Large-Scale Entanglement in Quantum Simulation

Trapped ions



UIBK & IQOQI

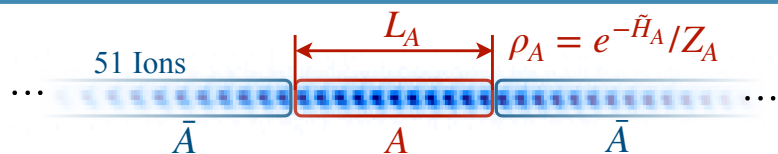
Theory: C Kokail, R van Bijnen, TV Zache, and P.Z.

Experiment: ML Joshi, F Kranzl, R Blatt, CF Roos

[Nature online Nov 23, 2023](#)

Early collaborations: M Dalmonte (→ ICTP), B Vermersch (→ Grenoble)

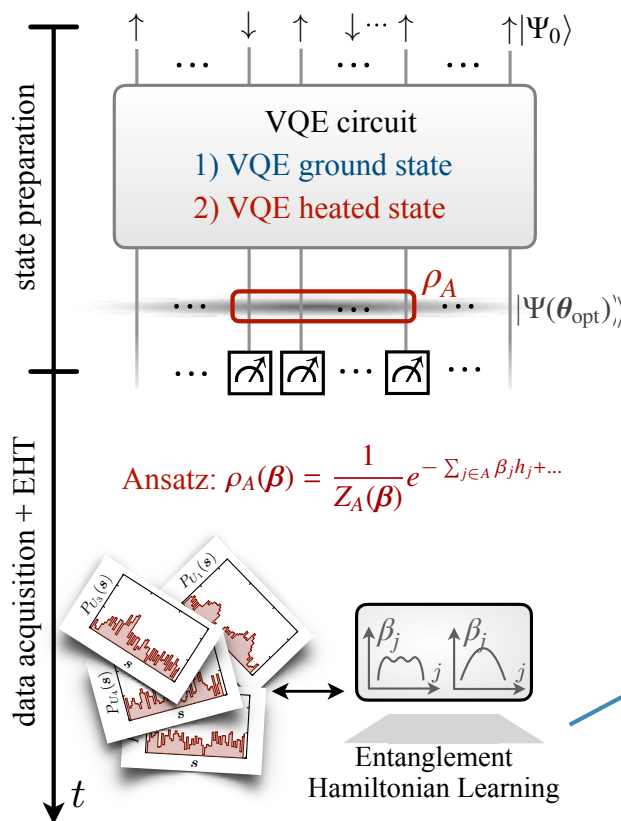
Heisenberg model



XXZ chain

$$H = J \sum_j (S_j^x S_{j+1}^x + S_j^y S_{j+1}^y + \Delta S_j^z S_{j+1}^z) = \sum_j h_j$$

State preparation & analysis



Entanglement properties

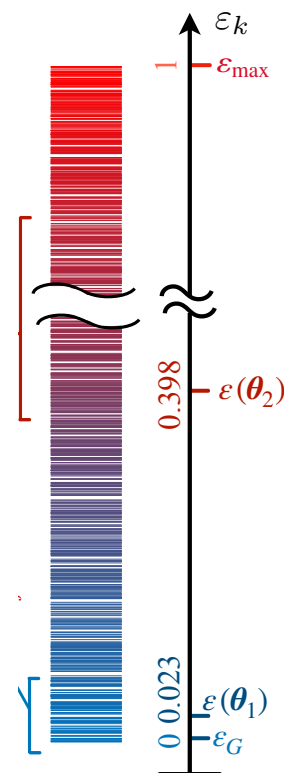


volume law entanglement $\longrightarrow$ heated state

sample-efficient tomography of ρ_A for subsystems > 20 lattice sites

area law entanglement $\longrightarrow$ ground state

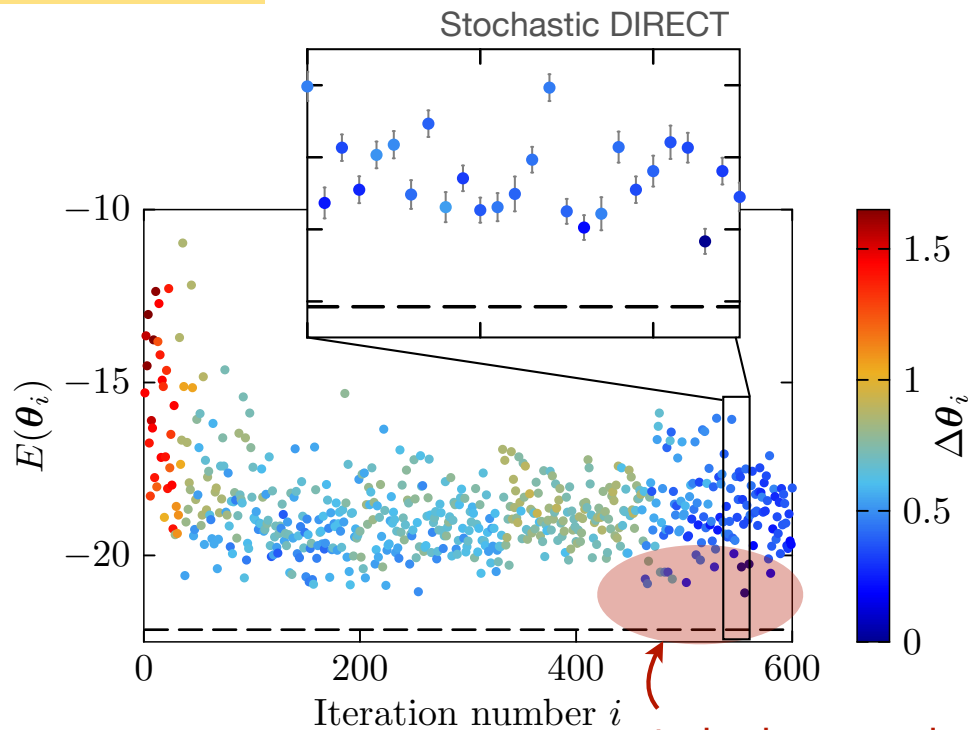
energy spectrum



Experimental Energy Optimization Trajectory for Ground State (VQE)

Theory: C Kokail, R van Bijnen et al., unpublished
Experiment: M Joshi et al., unpublished

N = 51 ions



to be improved

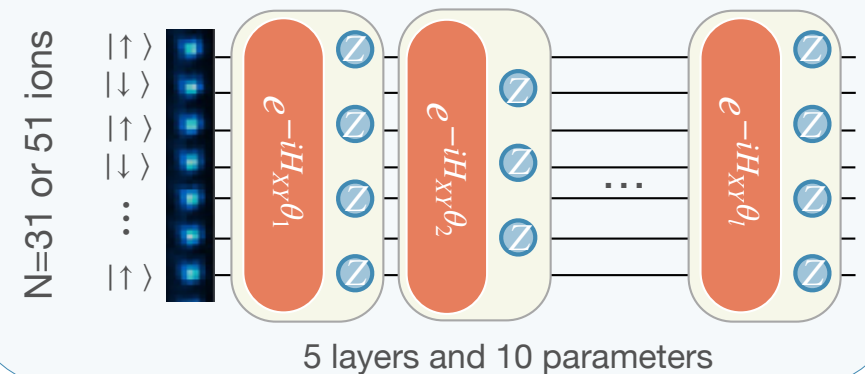
$\sim$ low temperature state $T \sim \text{few } J$

Heisenberg Model (spin- $\frac{1}{2}$)

$$\hat{H} = J \sum_{i=1}^{N-1} (\hat{S}_i^x \hat{S}_{i+1}^x + \hat{S}_i^y \hat{S}_{i+1}^y) + \Delta \sum_{i=1}^{N-1} \hat{S}_i^z \hat{S}_{i+1}^z + h \sum_{i=1}^N \hat{S}_i^z$$

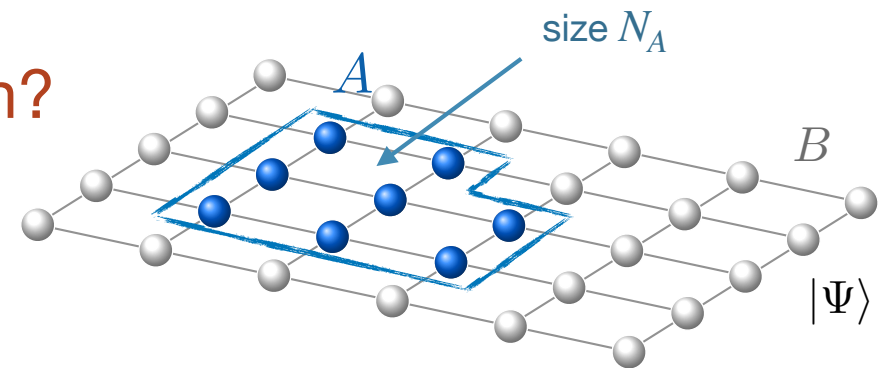
$J = 1 \quad \Delta = 1 \quad h = 0.5$

VQE Circuit with Trapped Ion Resources



preliminary

Learning the Entanglement Hamiltonian?



Protocol 0: Quantum state tomography

measurement
data



$$\rho_A = \exp(-\tilde{H}_A)$$

✓ expensive *

exponential in
subsystem size N_A

?

* except: for small system sizes, or if we know something about the quantum state

Do we know something about the *structure of $\tilde{H}_A$* to make tomography *efficient*?

A. Anshu et al., *Sample-Efficient Learning of Interacting Quantum Systems*, Nat. Phys. **17**, 931 (2021).

Entanglement Hamiltonian in QFT: *Bisognano-Wichmann Theorem*

Relativistic Quantum Field Theory

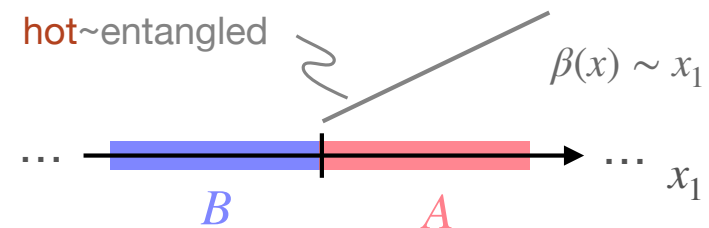
Lorentz invariance

$$H = \int_{A \cup B} d^d x \mathcal{H}(x)$$

vacuum state $|\Omega\rangle$



Entanglement Hamiltonian



$$\rho_A \equiv \text{Tr}_B |\Omega\rangle\langle\Omega| = \exp \left[- \int_A d^d x \beta(x) \mathcal{H}(x) \right]$$

$\tilde{H}_A$

Gibbs state with *local* temperature $\beta(x) \sim x_1$

EH $\tilde{H}_A$ as *deformed* system Hamiltonian

Bisognano and Wichmann, J. Math. Phys. (1976) Casini, Huerta & Myers, Journal of HEP (2011)

Review: M Dalmonte, V Eisler, M Falconi, B Vermersch, *Entanglement Hamiltonians - from field theory to lattice models & experiments*, Ann Phys 2022,534, 2200064

Entanglement Hamiltonian in QFT: *Bisognano-Wichmann Theorem*

Conformal Field Theory

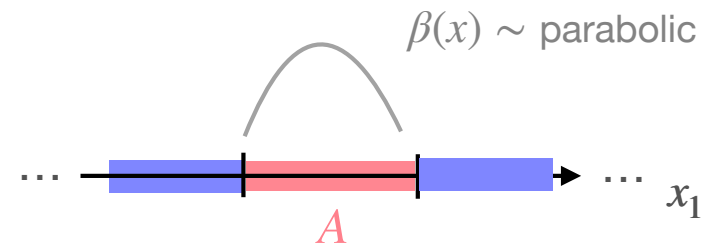
scale invariance

$$H = \int_{A \cup B} d^d x \mathcal{H}(x)$$

vacuum state $|\Omega\rangle$



Entanglement Hamiltonian



$$\rho_A \equiv \text{Tr}_B |\Omega\rangle\langle\Omega| = \exp \left[- \int_A d^d x \beta(x) \mathcal{H}(x) \right]$$

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Gibbs state with *local* temperature $\beta(x) \sim x_1$

Entanglement Hamiltonian as *deformed* system Hamiltonian

Casini, Huerta & Myers, Journal of HEP (2011)

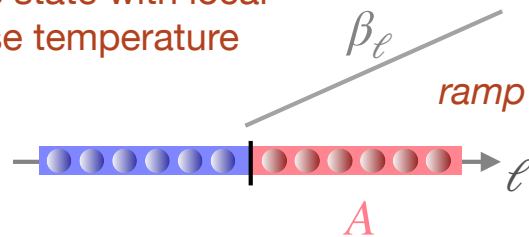
1+1 CFT: Hislop, Longo, Cardy, Calabrese, Tonni, Wen, Ryu, Ludwig, ... (ground state, thermal & quench)

Review: M Dalmonte, V Eisler, M Falconi, B Vermersch, *Entanglement Hamiltonians - from field theory to lattice models & experiments*, Ann Phys 2022,534, 2200064

Lattice Bisognano-Wichmann & beyond

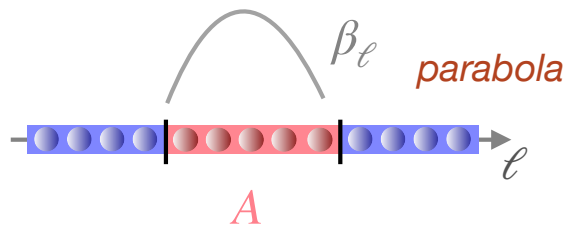
ground state of many-body lattice model

Gibbs state with local
inverse temperature



$$\rho_A = e^{-\tilde{H}_A}$$

entanglement
Hamiltonian



BW recipe

$$\hat{H} = \sum_{\ell} \hat{h}_{\ell} \quad \text{k-local Hamiltonian}$$

$$\tilde{H}_A = \sum_{\ell \in A} \beta_{\ell} \hat{h}_{\ell} + \dots \quad \text{EH as local deformation of system Hamiltonian}$$

1. Validity of BW-like EH, *non-local corrections*
often sub-leading

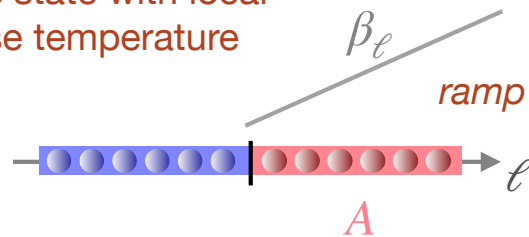
Analytical results: non-interacting, non-critical chains

Numerical evidence: lattice models, quench dynamics

Lattice Bisognano-Wichmann & beyond

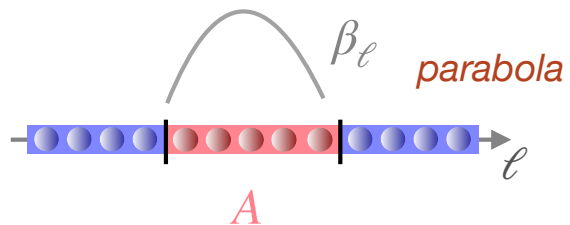
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entanglement
Hamiltonian

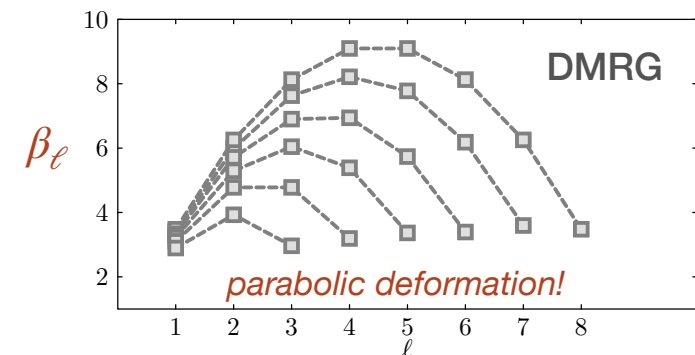


BW recipe

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$$\tilde{H}_A = \sum_{\ell \in A} \beta_{\ell} \hat{h}_{\ell} + \dots \quad \text{EH as local deformation of system Hamiltonian}$$

Numerical example: Heisenberg model 1D

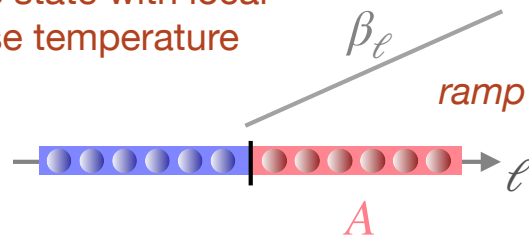


Review: M Dalmonte, V Eisler, M Falconi, B Vermersch, *Entanglement Hamiltonians - from field theory to lattice models & experiments*, Ann Phys 2022,534, 2200064

Lattice Bisognano-Wichmann & beyond

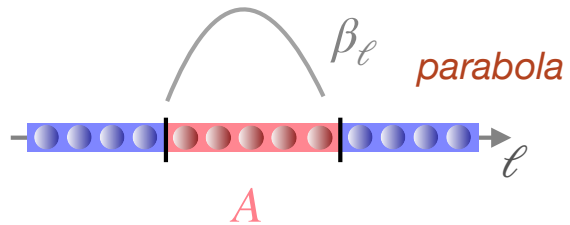
ground state of many-body lattice model

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$$\rho_A = e^{-\tilde{H}_A}$$

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$$\tilde{H}_A = \sum_{\ell \in A} \beta_{\ell} \hat{h}_{\ell} + \dots \quad \text{EH as local deformation of system Hamiltonian}$$

2. suggests an efficient ansatz
to 'learn' Entanglement Hamiltonian

Review: M Dalmonte, V Eisler, M Falconi, B Vermersch, *Entanglement Hamiltonians - from field theory to lattice models & experiments*, Ann Phys 2022,534, 2200064

Entanglement in Many-Body Quantum Systems

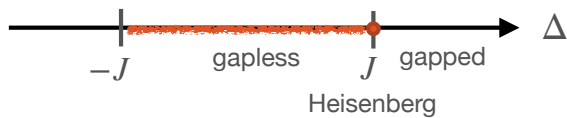
Many-Body Problem

Hamiltonian

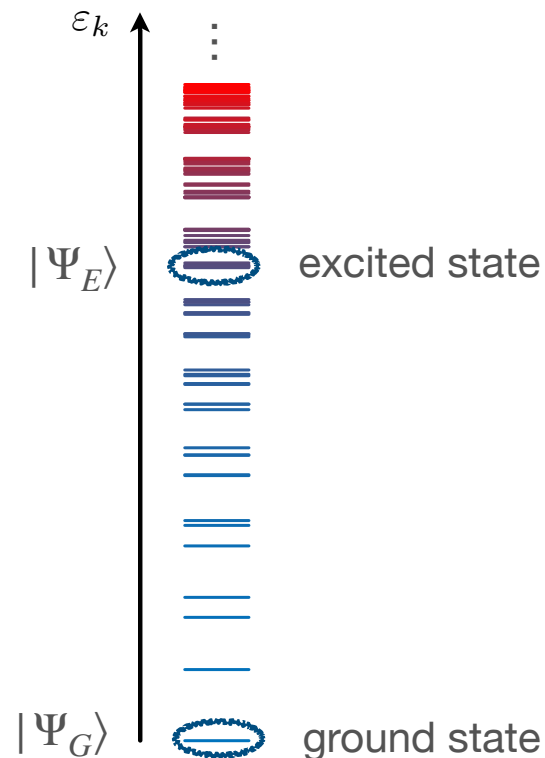
$$\hat{H} = \sum_j \hat{h}_j \quad \text{k-local}$$

Example: XXZ / Heisenberg model (1D)

$$\hat{H}_T = J \sum_{i=1}^{N-1} \left(\hat{S}_i^x \hat{S}_{i+1}^x + \hat{S}_i^y \hat{S}_{i+1}^y \right) + \Delta \sum_{i=1}^{N-1} \hat{S}_i^z \hat{S}_{i+1}^z$$



energy spectrum



excited state

$$S_A \propto V = L_A^d$$

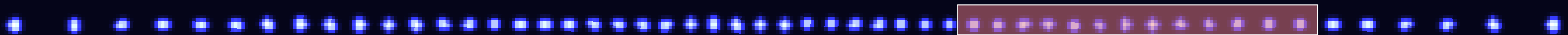
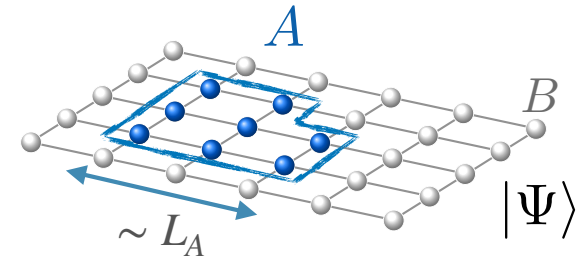
thermal entropy

volume law entanglement

$$S_A \propto L_A^{d-1}$$

area law entanglement

$$\sim c \log L_A \text{ CFT } d = 1$$

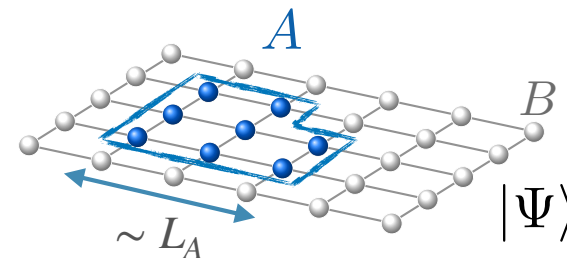


Area Law vs. Volume Law Entanglement

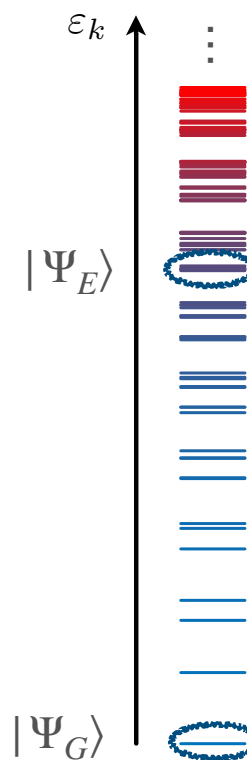
Many-Body Problem

Hamiltonian

$$\hat{H} = \sum_j \hat{h}_j \quad \text{k-local}$$



energy spectrum



q-simulators can represent volume law states $\rightarrow$ quantum advantage

excited state

$$S_A \propto V = L_A^d$$

thermal entropy

volume law entanglement

classical simulations with tensor networks

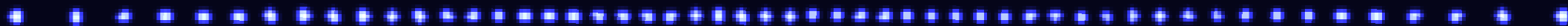
$|\Psi_G\rangle$

ground state

$$S_A \propto L_A^{d-1}$$

area law entanglement

$$\sim c \log L_A \text{ CFT } d = 1$$



Entanglement Hamiltonian & Eigenstate Thermalization Hypothesis

Eigenstate Thermalization Hypothesis (ETH):

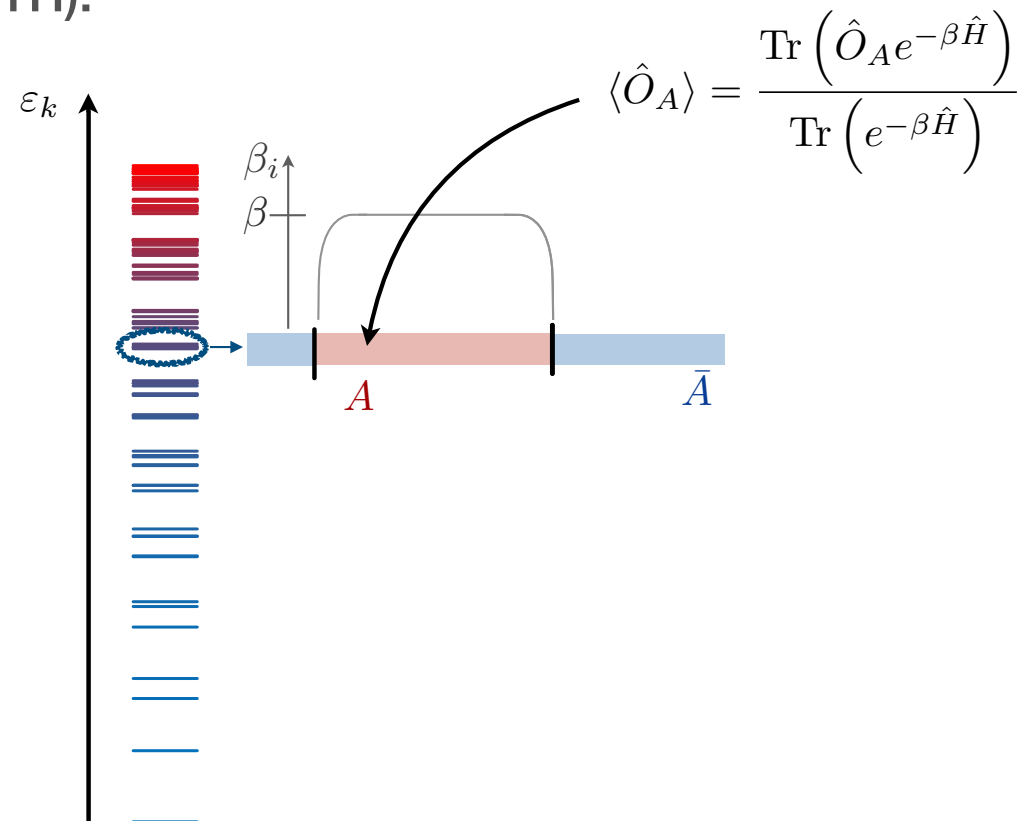
local Hamiltonian: $\hat{H} = \sum_i \hat{h}_i$

reduced density matrix of a subsystem A

$$\rho_A \approx \frac{1}{Z} e^{-\beta \hat{H}_A}$$

$$\approx \frac{1}{Z} e^{-\sum_i \beta_i \hat{h}_i}$$

local temperature profile



Entanglement Hamiltonian & Eigenstate Thermalization Hypothesis

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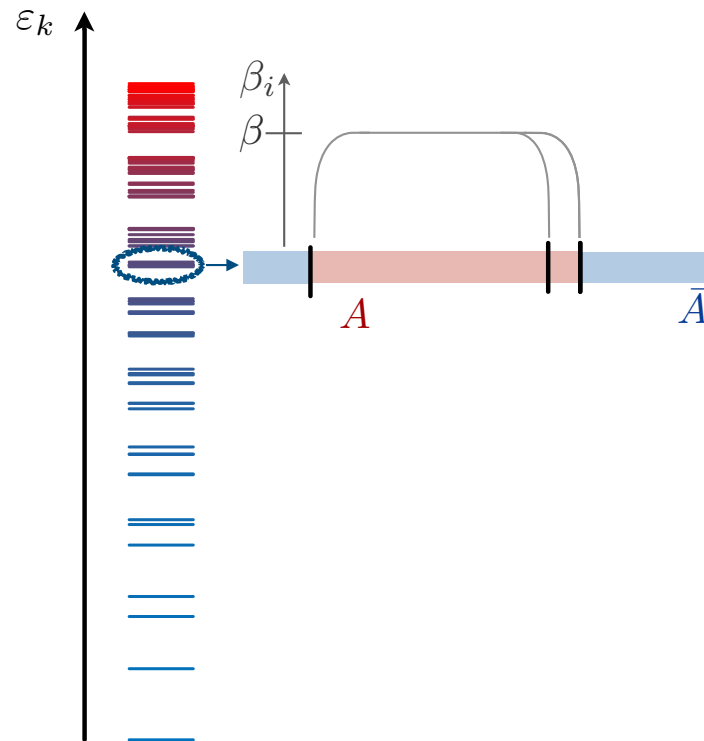
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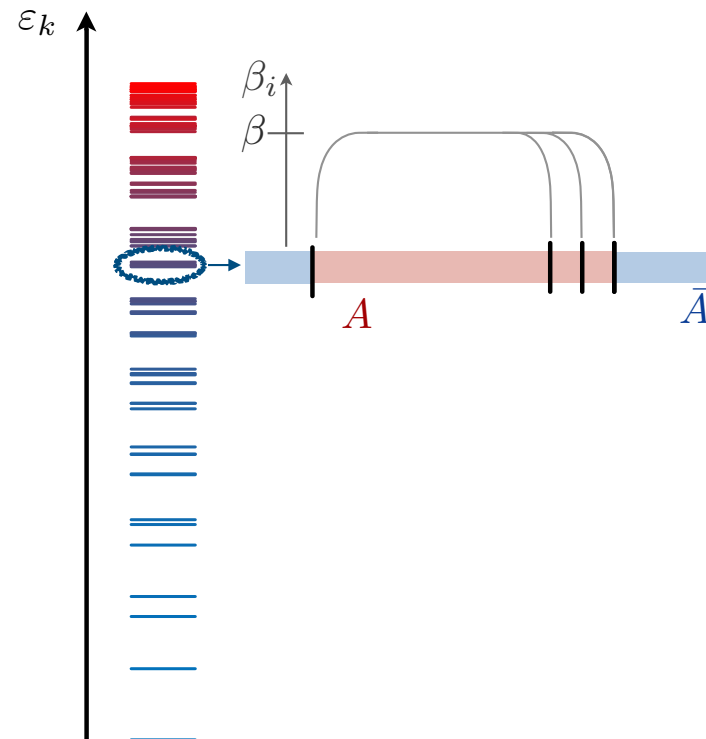
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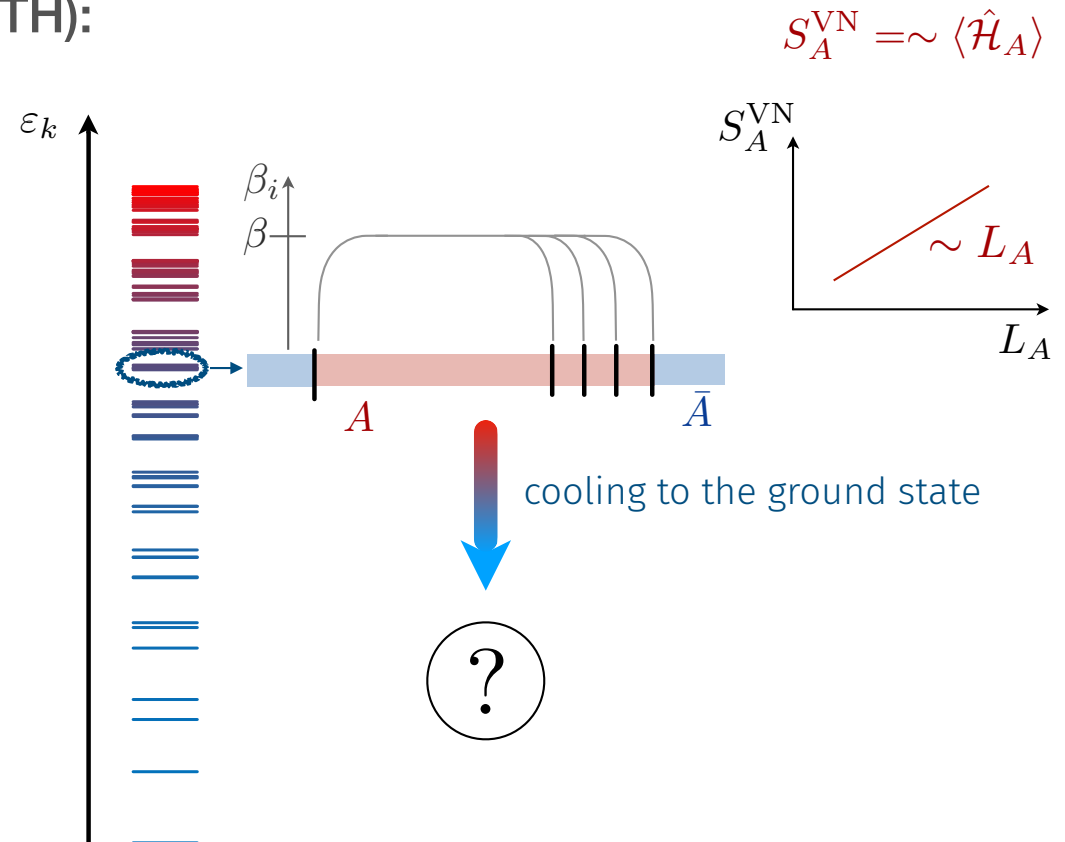
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Entanglement Hamiltonian & Eigenstate Thermalization Hypothesis

EH from CFT for ground state:
(at the critical point)

$$\rho_A = \frac{1}{Z} \exp \left[- \int_A dx \beta(x) \mathcal{H}(x) \right]$$

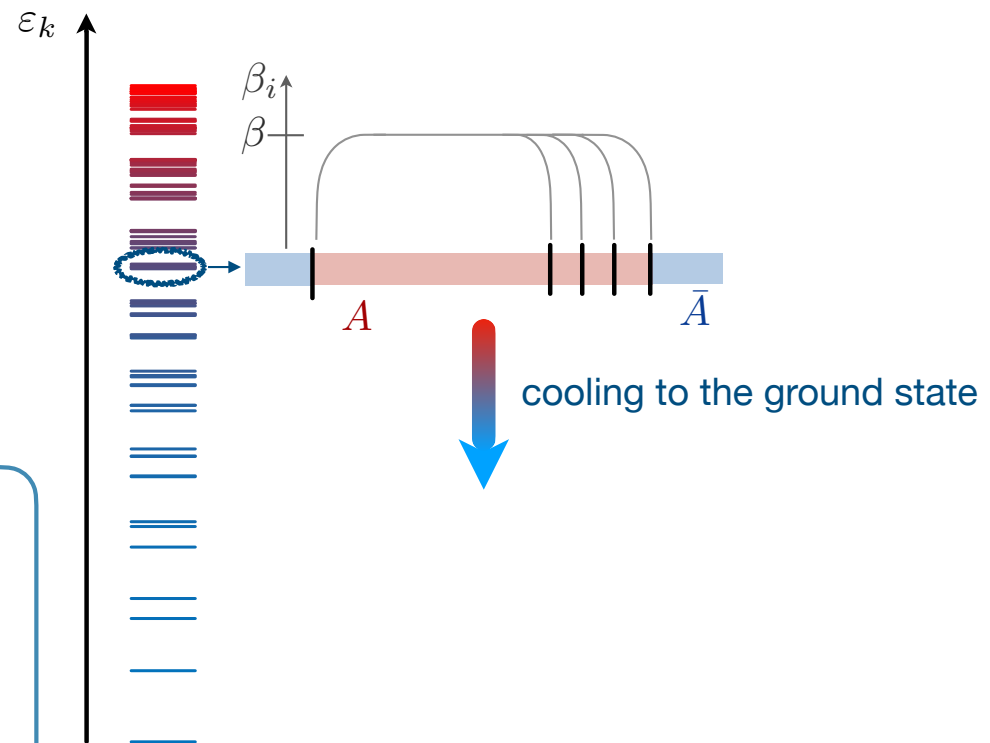
with

$$\beta(x) \propto \frac{L_A^2 - 4x^2}{L_A}$$

Parametrization as Gibbs state with *local inverse temperature*

$$\rho_A \sim e^{-\sum_{i \in A} \beta_i \hat{h}_i}$$

local temperature



Entanglement Hamiltonian & Eigenstate Thermalization Hypothesis

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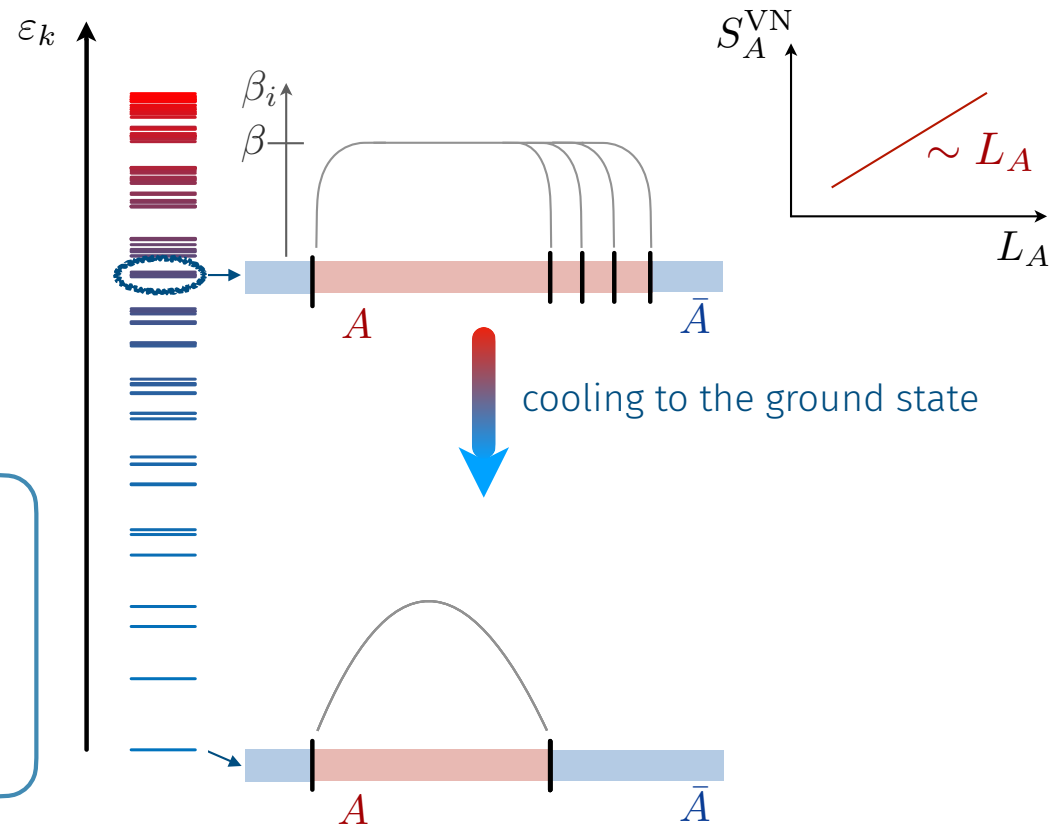
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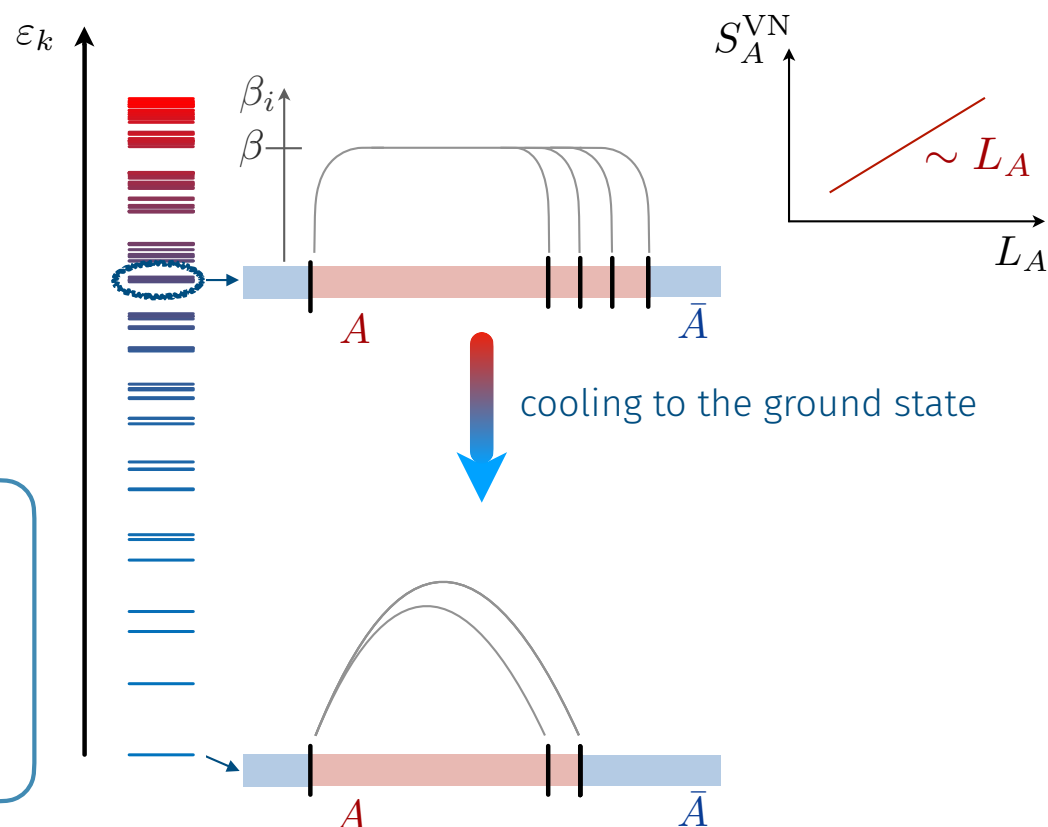
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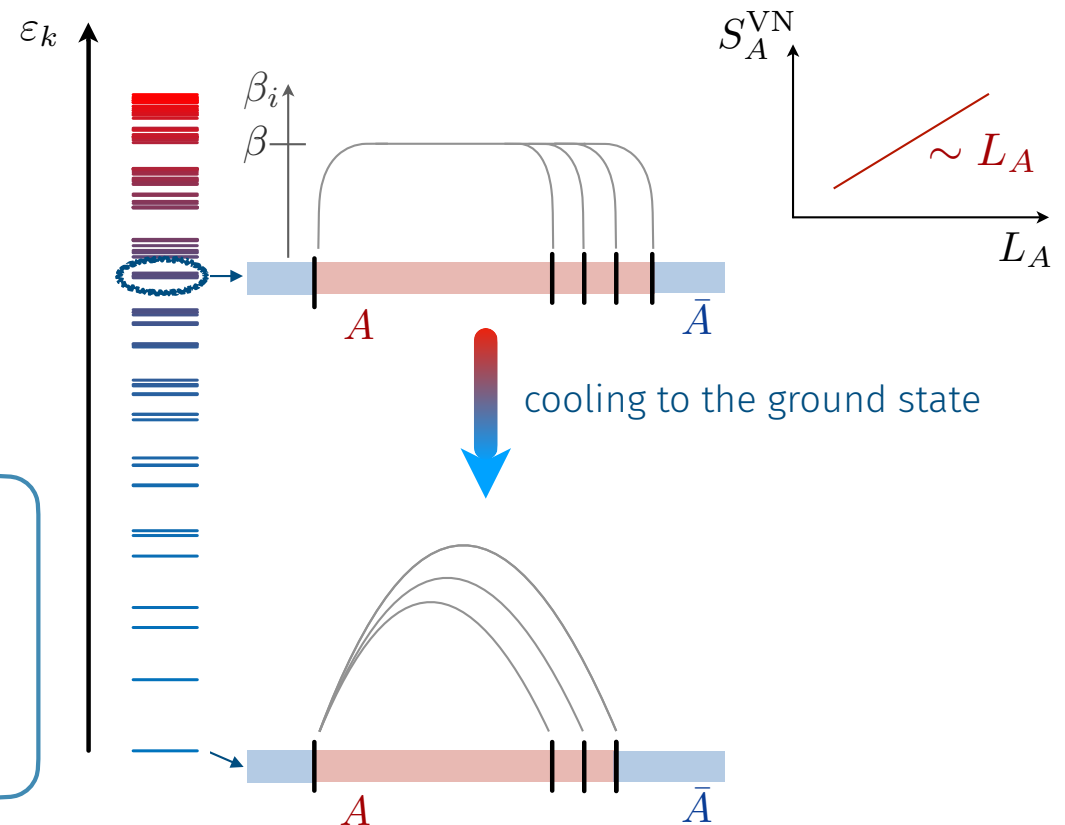
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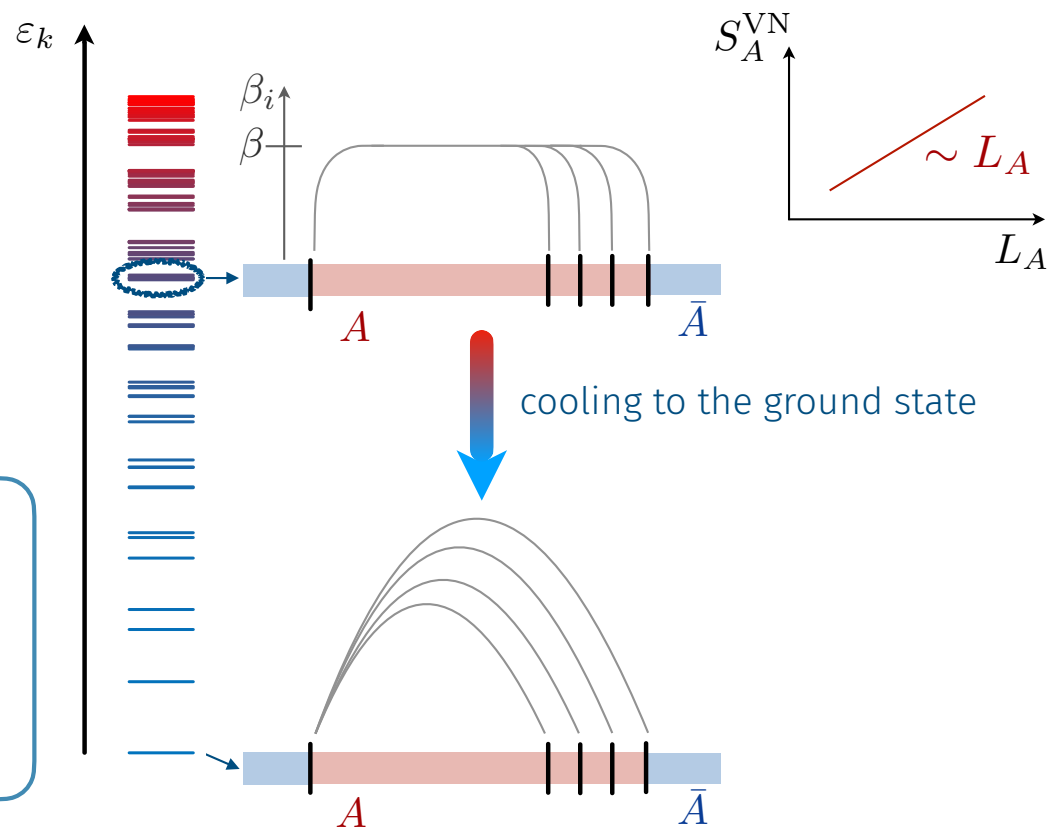
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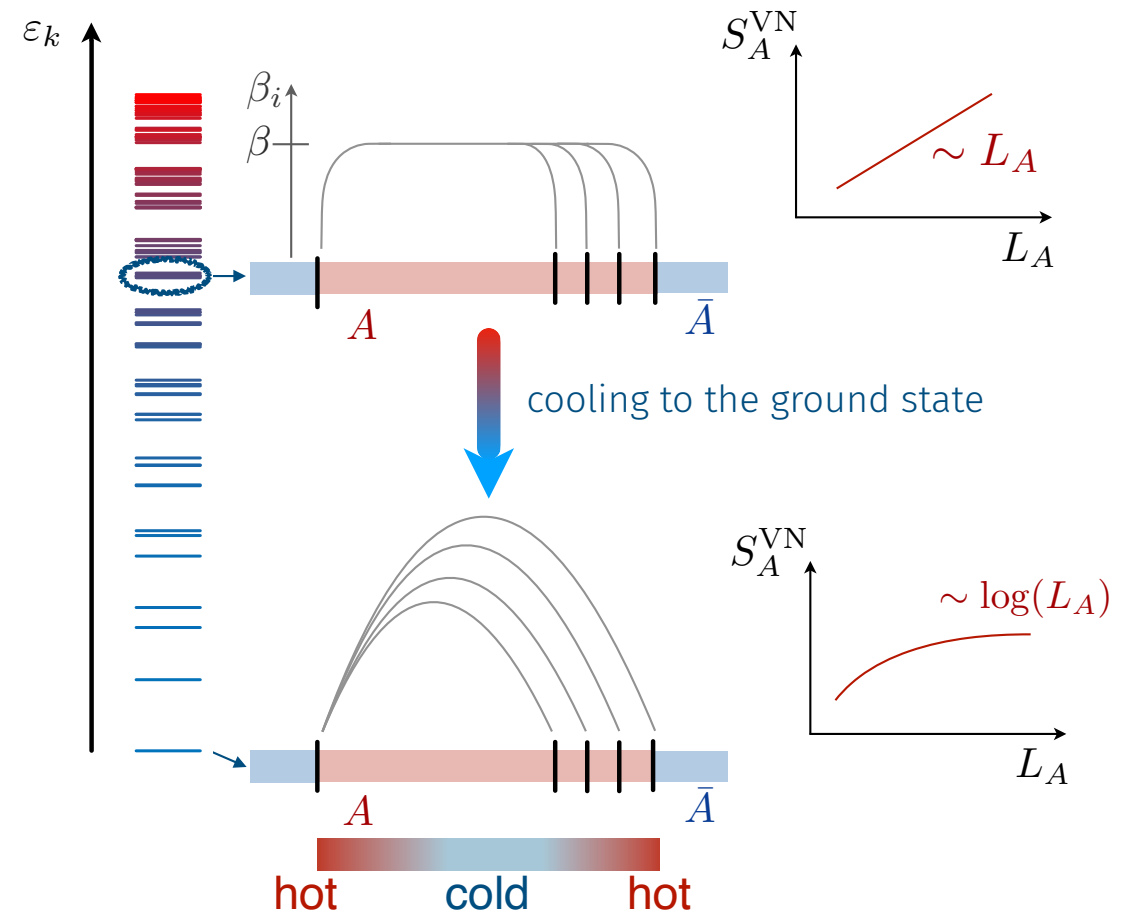
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local temperature profile



Entanglement Hamiltonian & Eigenstate Thermalization Hypothesis

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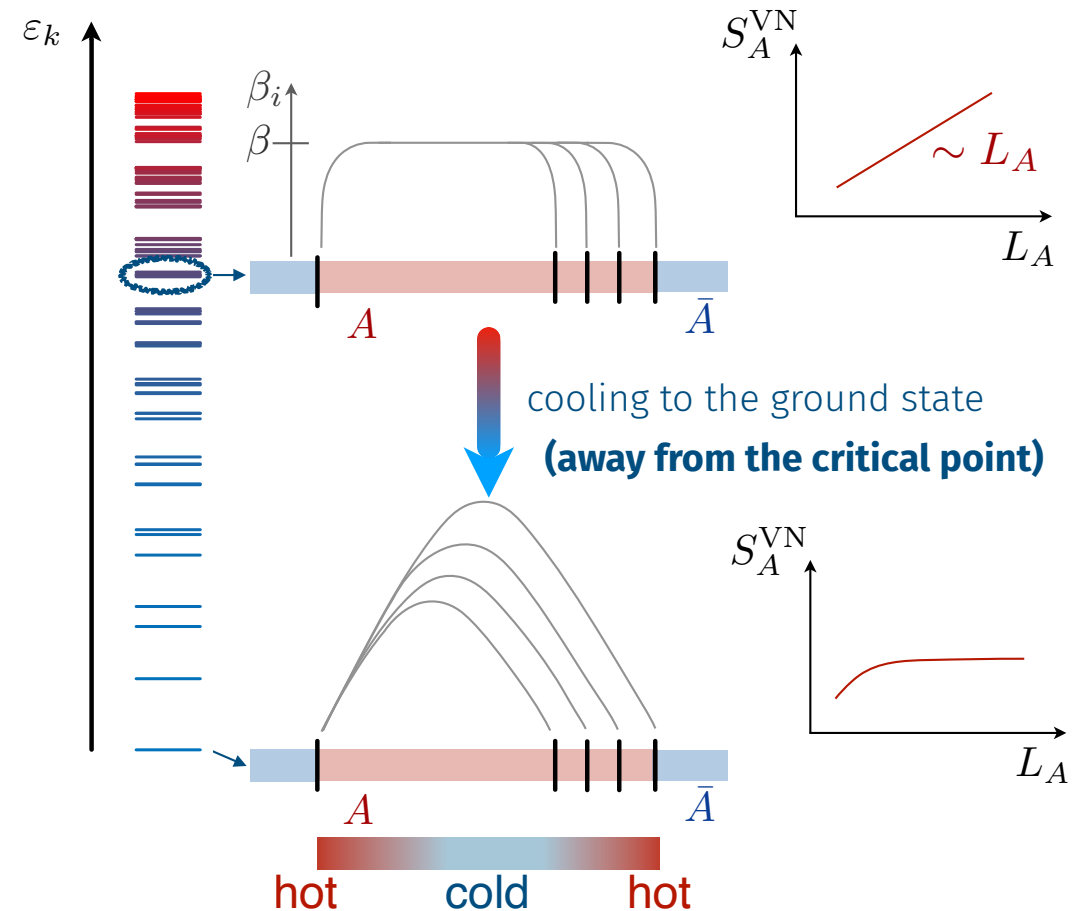
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local temperature profile



Viktor Eisler *et al* *J. Stat. Mech.* (2020) 103102

Casini, Huerta & Myers, *JHEP*, 2011(5), 1-41

Entanglement Hamiltonian & Eigenstate Thermalization Hypothesis

Example: XXZ model (51 sites)

$$\hat{H} = \frac{1}{2} \sum_i \left(\hat{S}_i^+ \hat{S}_{i+1}^- + \text{H.c.} \right) + \Delta \sum_i \hat{S}_i^z \hat{S}_{i+1}^z$$

Ansatz for the Entanglement Hamiltonian:

ferromagnet c=1 CFT antiferromagnet

$$\hat{\mathcal{H}}_A = \sum_{\ell} \beta_{\Delta \ell} \hat{h}_{\ell} - 1 \quad \Delta = 1$$

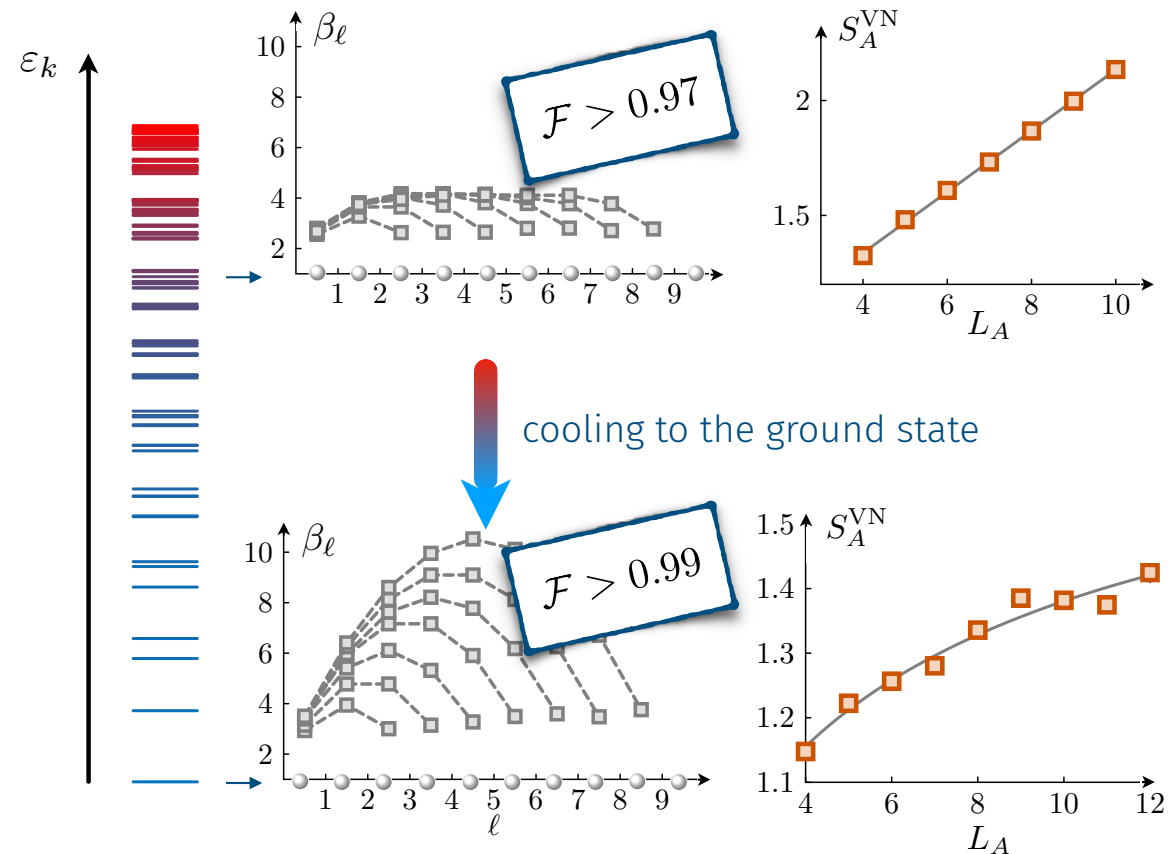
$$\hat{h}_{\ell} = \frac{1}{2} \left(\hat{S}_{\ell}^+ \hat{S}_{\ell+1}^- + \text{H.c.} \right) + \Delta \hat{S}_{\ell}^z \hat{S}_{\ell+1}^z$$

Qi & Ranard Quantum 3, 159 (2019)

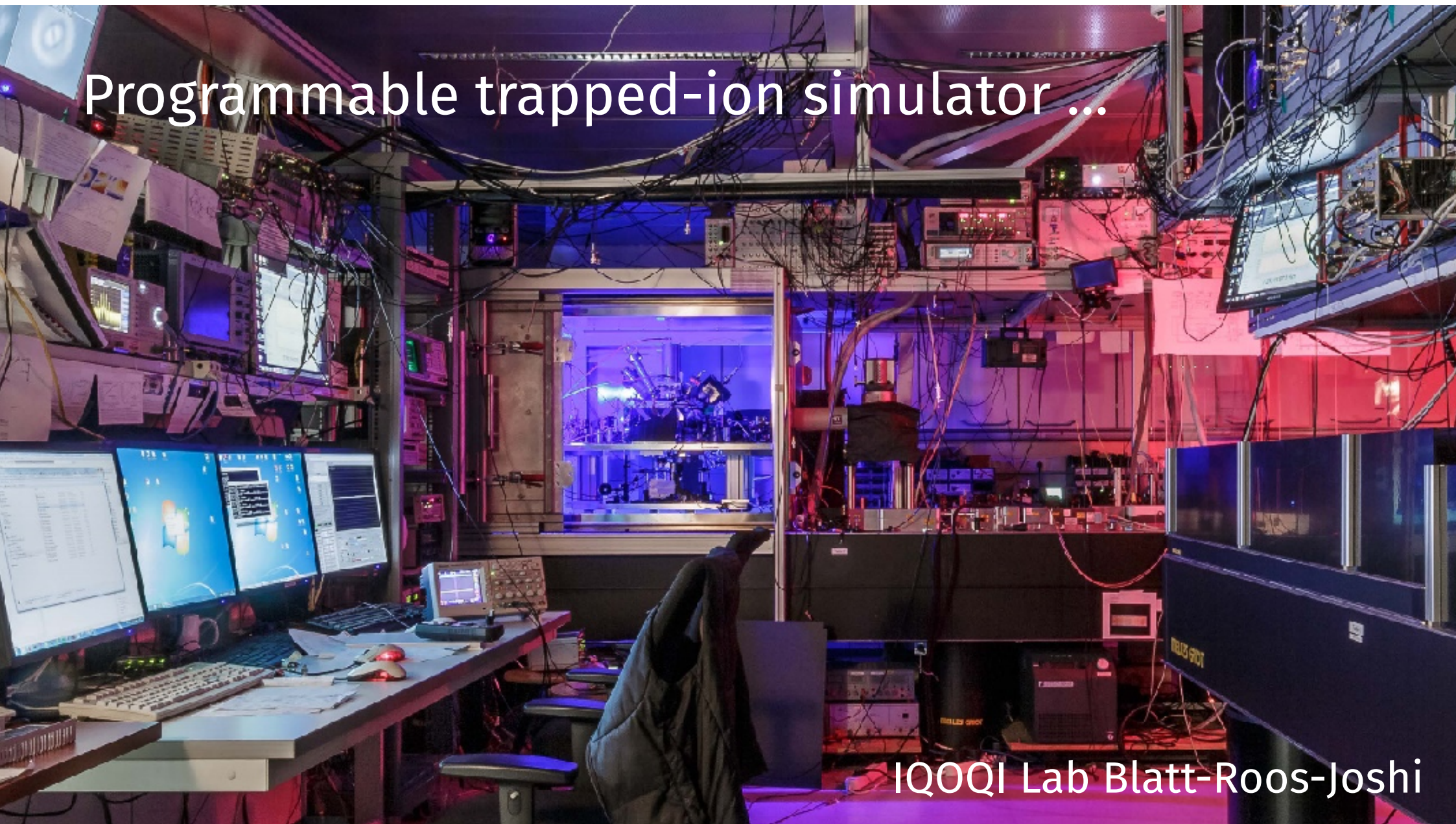
E. Bairey et.al. Phys. Rev. Lett. (2019)

W. Zhu et.al. Phys. Rev. B 99, 235109

C. Kokail et.al. Nat.Phys. 2021



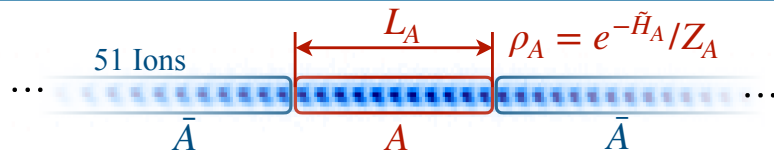
Programmable trapped-ion simulator ...



IQOQI Lab Blatt-Roos-Joshi

Results: Theory and Experiment

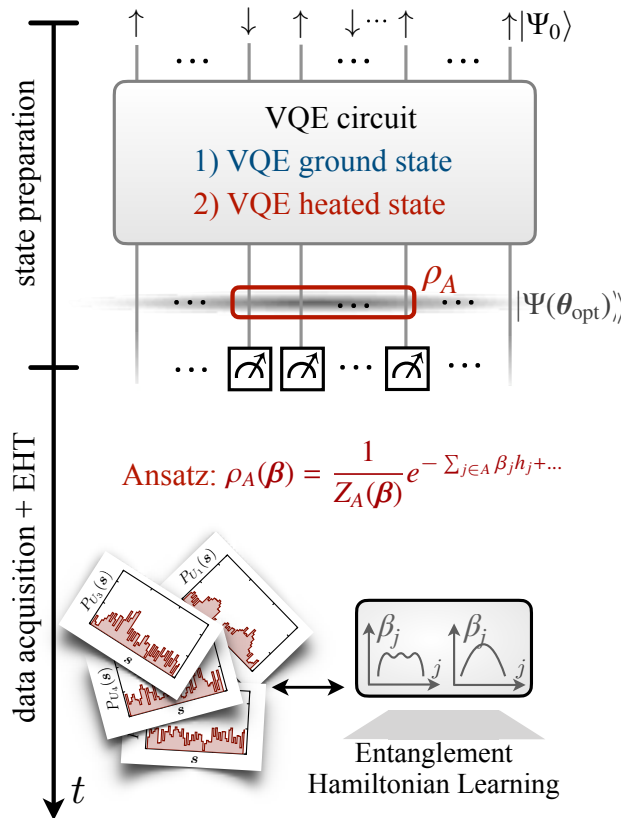
Heisenberg model



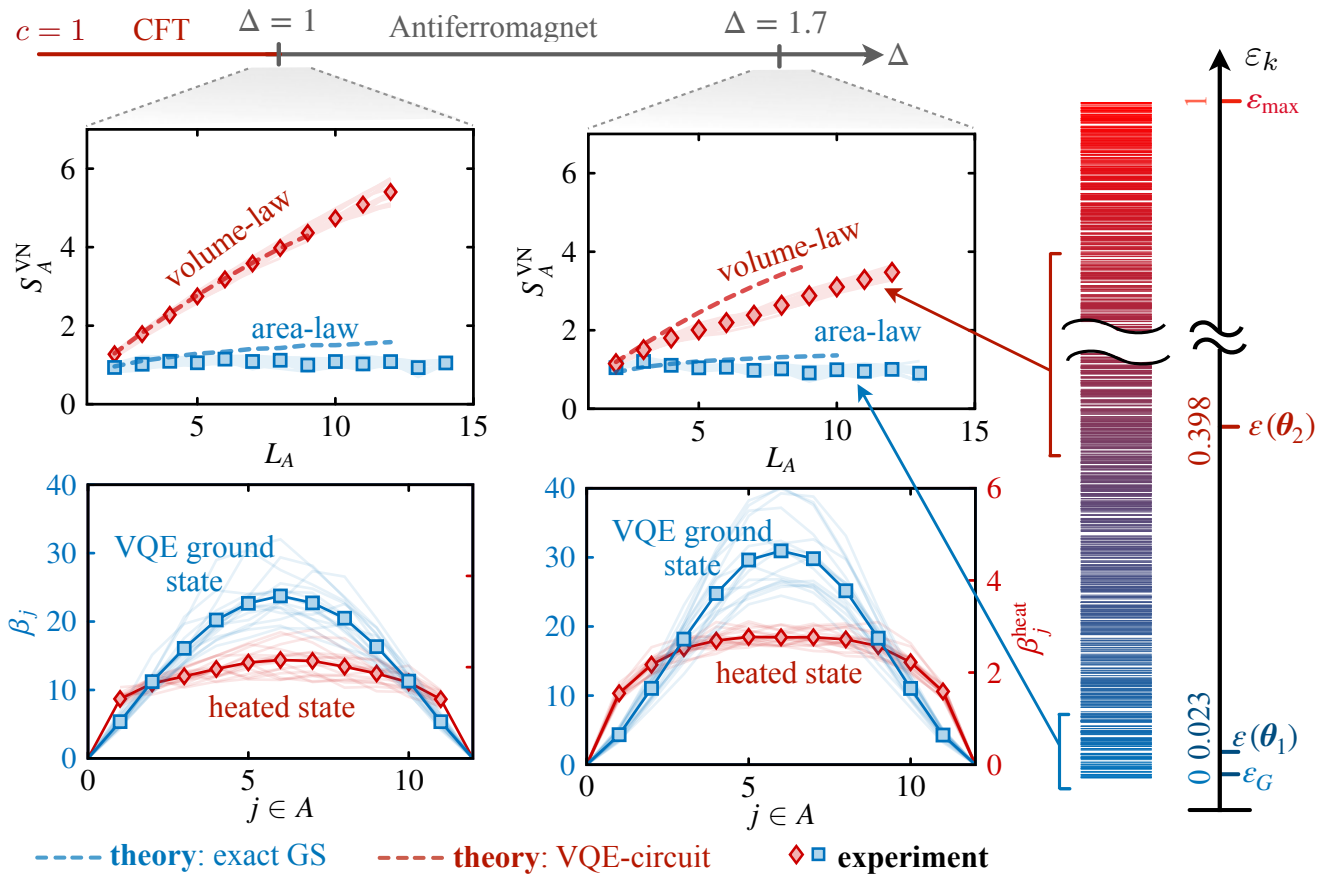
XXZ chain

$$H = J \sum_j (S_j^x S_{j+1}^x + S_j^y S_{j+1}^y + \Delta S_j^z S_{j+1}^z) = \sum_j h_j$$

State preparation & analysis

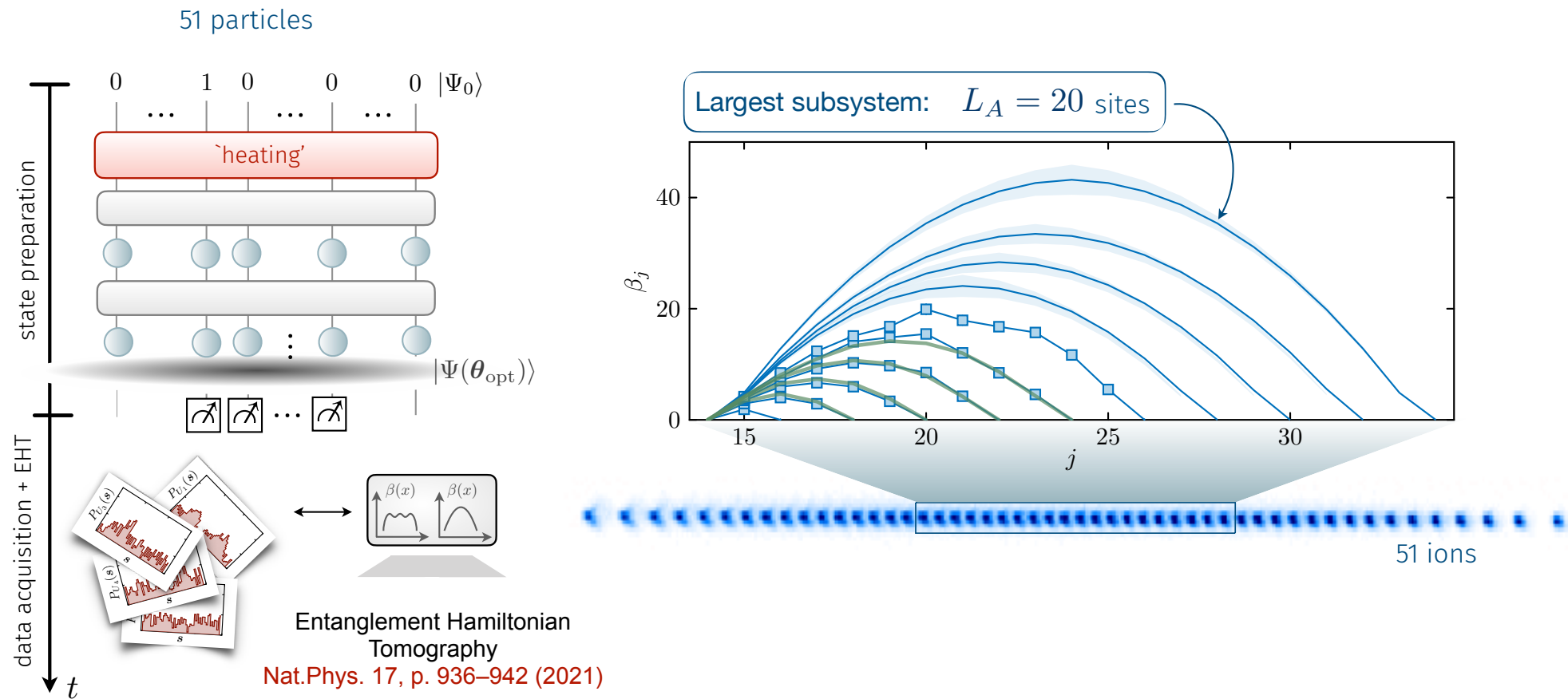


Entanglement properties



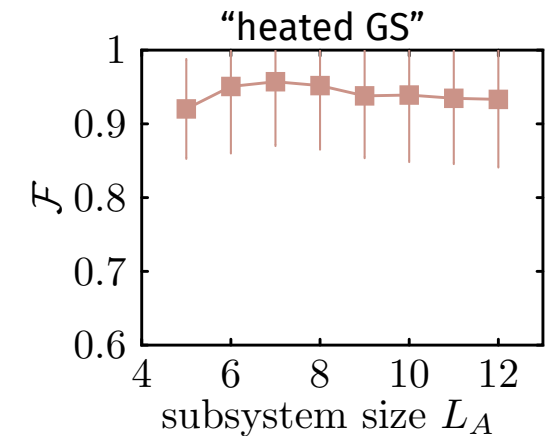
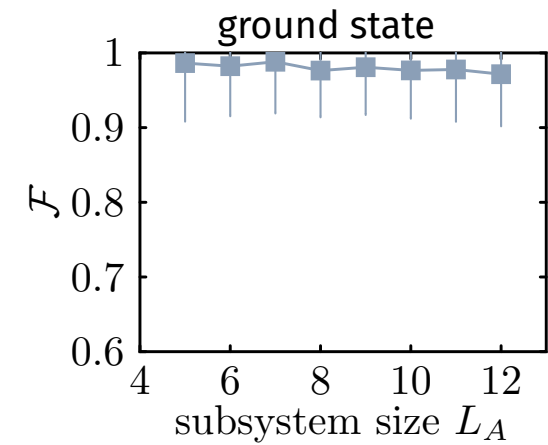
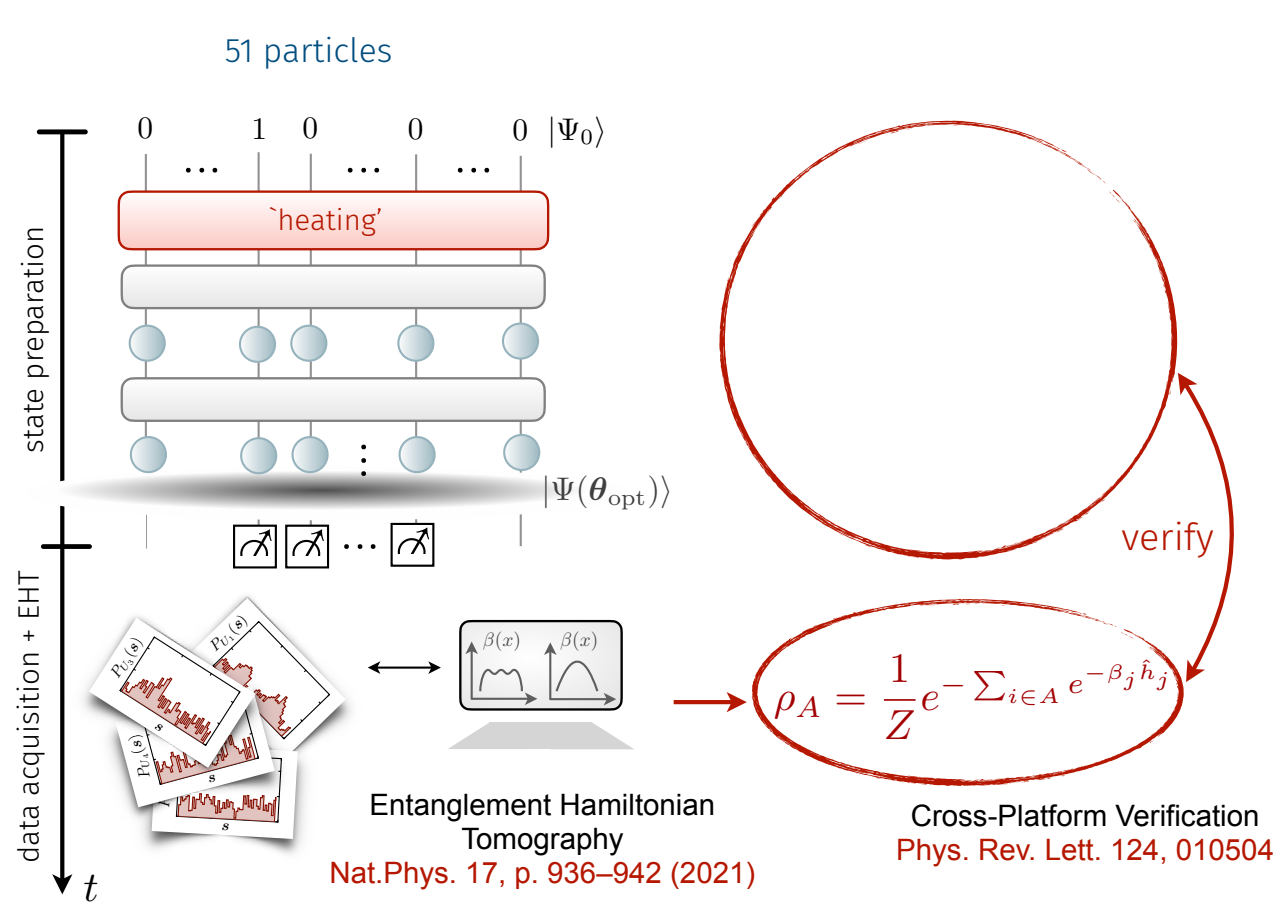
Scaling

variational prep. of the XXZ GS:
$$\hat{H} = J \sum_{j=1}^L \left(\hat{S}_j^x \hat{S}_{j+1}^x + \hat{S}_j^y \hat{S}_{j+1}^y + \Delta \hat{S}_j^z \hat{S}_{j+1}^z \right) = \sum_{j=1}^L \hat{h}_j \quad L = 51$$



Verification of Entanglement Hamiltonian Tomography

variational prep. of the XXZ GS: $\hat{H} = J \sum_{j=1}^L \left(\hat{S}_j^x \hat{S}_{j+1}^x + \hat{S}_j^y \hat{S}_{j+1}^y + \Delta \hat{S}_j^z \hat{S}_{j+1}^z \right) = \sum_{j=1}^L \hat{h}_j \quad L = 51$



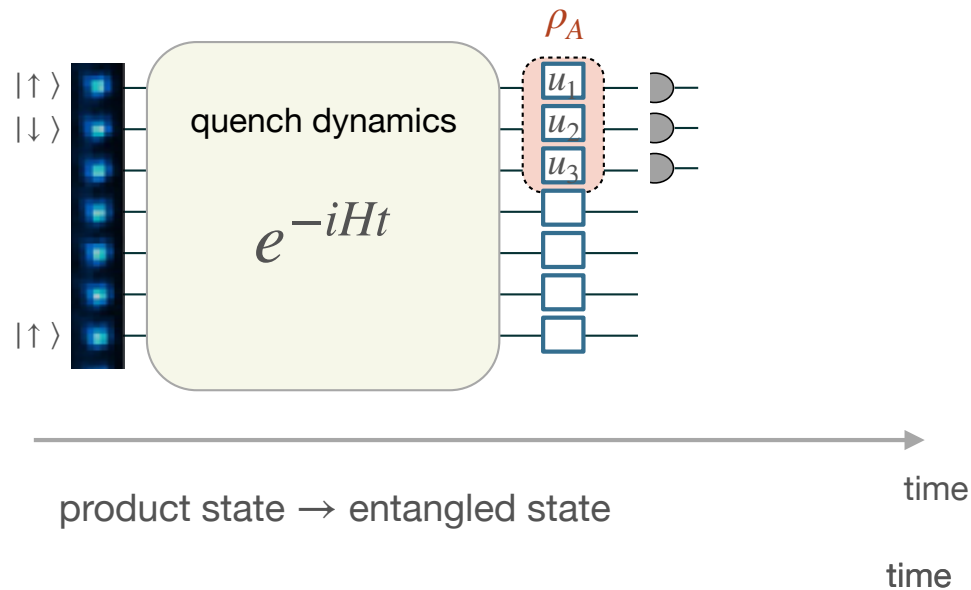
Details of *Entanglement Hamiltonian Learning*

C. Kokail, R. van Bijnen, A. Elben, B. Vermersch, & PZ, *Entanglement Hamiltonian Tomography in Quantum Simulation*, Nat. Phys. (2021).

A. Anshu, S. Arunachalam, T. Kuwahara, and M. Soleimanifar, *Sample-Efficient Learning of Interacting Quantum Systems*, Nat. Phys. (2021).

Randomized Measurements: Tomography

Quench dynamics with analog quantum simulator



Randomized Tomography

$$\rho_A = \mathbb{E}_{U \sim \text{CUE}}[\hat{\rho}_A]$$

exponentially expensive

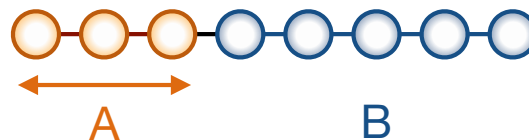


$$\hat{\rho}_A = \sum_{\mathbf{s}, \mathbf{s}'} \sum_U P_U(\mathbf{s}) (-2)^{-D[\mathbf{s}, \mathbf{s}']} U |\mathbf{s}'\rangle \langle \mathbf{s}'| U^\dagger$$

tomographically complete

Measuring (Large-Scale) Entanglement

Protocol 0: Quantum State Tomography



data



ρ_A

✓ expensive* $\sim \text{rank}(\rho_A) 2^{N_A}$ (scales exponentially)

sample-efficient entanglement
Hamiltonian tomography

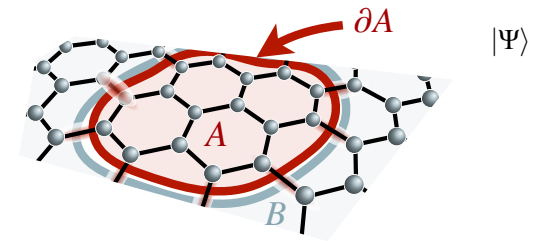
* tomography can be made 'more efficient' if we *know* something about the quantum state: MPS, low rank, neural network, ...

Classical Shadows

* or, we are only interested in certain functionals of ρ_A , e.g. expectation values $\langle O \rangle = \text{Tr } O_A \rho_A$

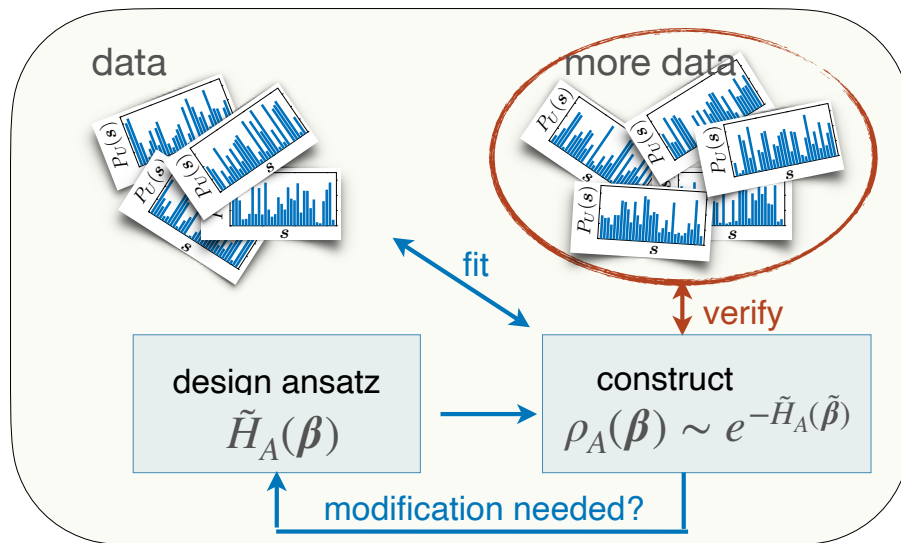
Sample-Efficient Learning of the Entanglement Hamiltonian (EH)

Protocol: *efficient* parametrization of EH



data $\longrightarrow$ $\rho_A = e^{-\tilde{H}_A(\beta)}$ with $\tilde{H}_A(\beta) = \sum_{i \in A} \beta_i \hat{h}_\ell + \dots$ polynomial # β_i ?

Gibbs state simple operator structure



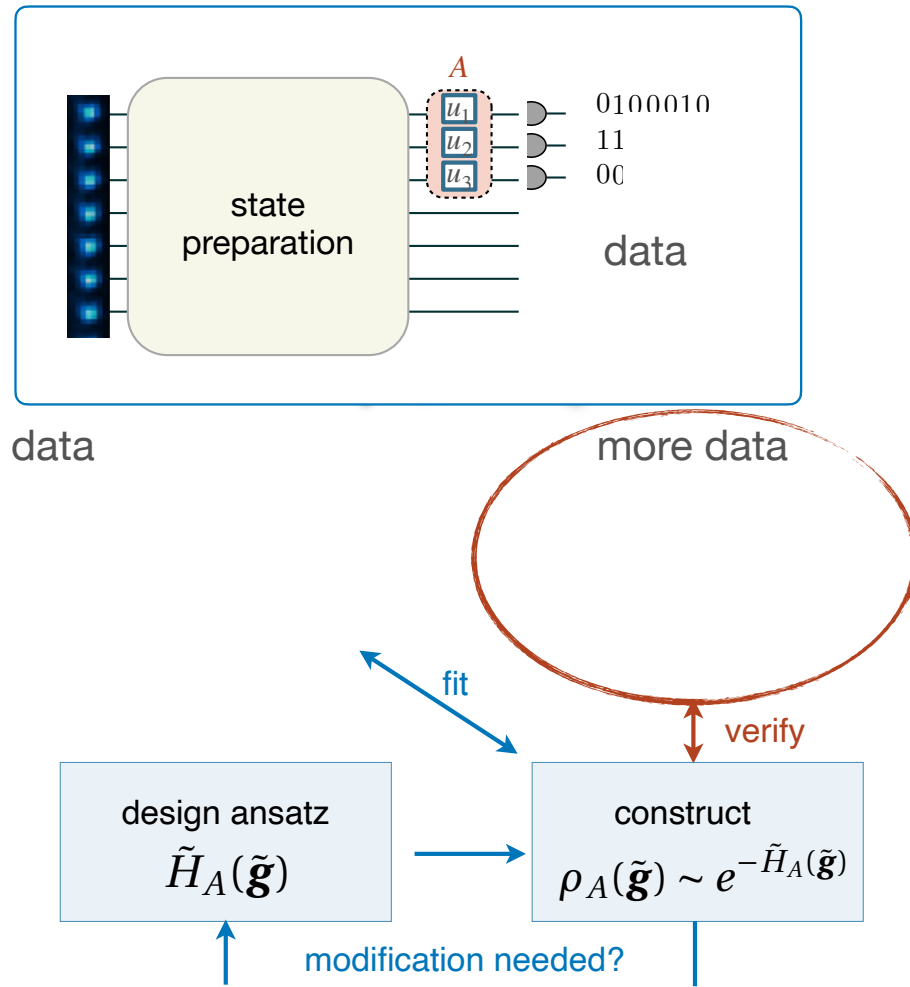
measurement protocol

- sample complexity
- time efficiency

C. Kokail, R. van Bijnen, A. Elben, B. Vermersch, & PZ, *Entanglement Hamiltonian Tomography in Quantum Simulation*, Nat. Phys. (2021).

A. Anshu, S. Arunachalam, T. Kuwahara, and M. Soleimanifar, *Sample-Efficient Learning of Interacting Quantum Systems*, Nat. Phys. (2021).

Learning the Entanglement Hamiltonian



Learning Protocol

- Measure experimental frequencies:

$$P_U(\mathbf{s}) = \text{Tr} \left(U \rho_A U^\dagger |\mathbf{s}\rangle \langle \mathbf{s}| \right)$$

- Ansatz for Entanglement Hamiltonian:

$\tilde{H}_A(\tilde{\mathbf{g}})$ • e.g. deformation of system Hamiltonian plus corrections

- Fit optimal parameters $\tilde{\mathbf{g}}$ by minimizing the distance to the frequencies

$$\chi^2(\tilde{\mathbf{g}}) = \sum_{U, \mathbf{s}} \left[\text{Tr} \left(U |\mathbf{s}\rangle \langle \mathbf{s}| U^\dagger \frac{e^{-\tilde{H}_A(\tilde{\mathbf{g}})}}{Z(\tilde{\mathbf{g}})} \right) - P_U(\mathbf{s}) \right]^2$$

- Verify by measuring Hilbert-Schmidt fidelities

$$\mathcal{F} \sim \text{Tr} [\rho_A^{\text{data}} \rho_A^{\text{more data}}] = \dots$$

A. Elben et. al. PRL (2020)

EHT for the Ground state of a long-range Ising chain (Theory)

System Hamiltonian

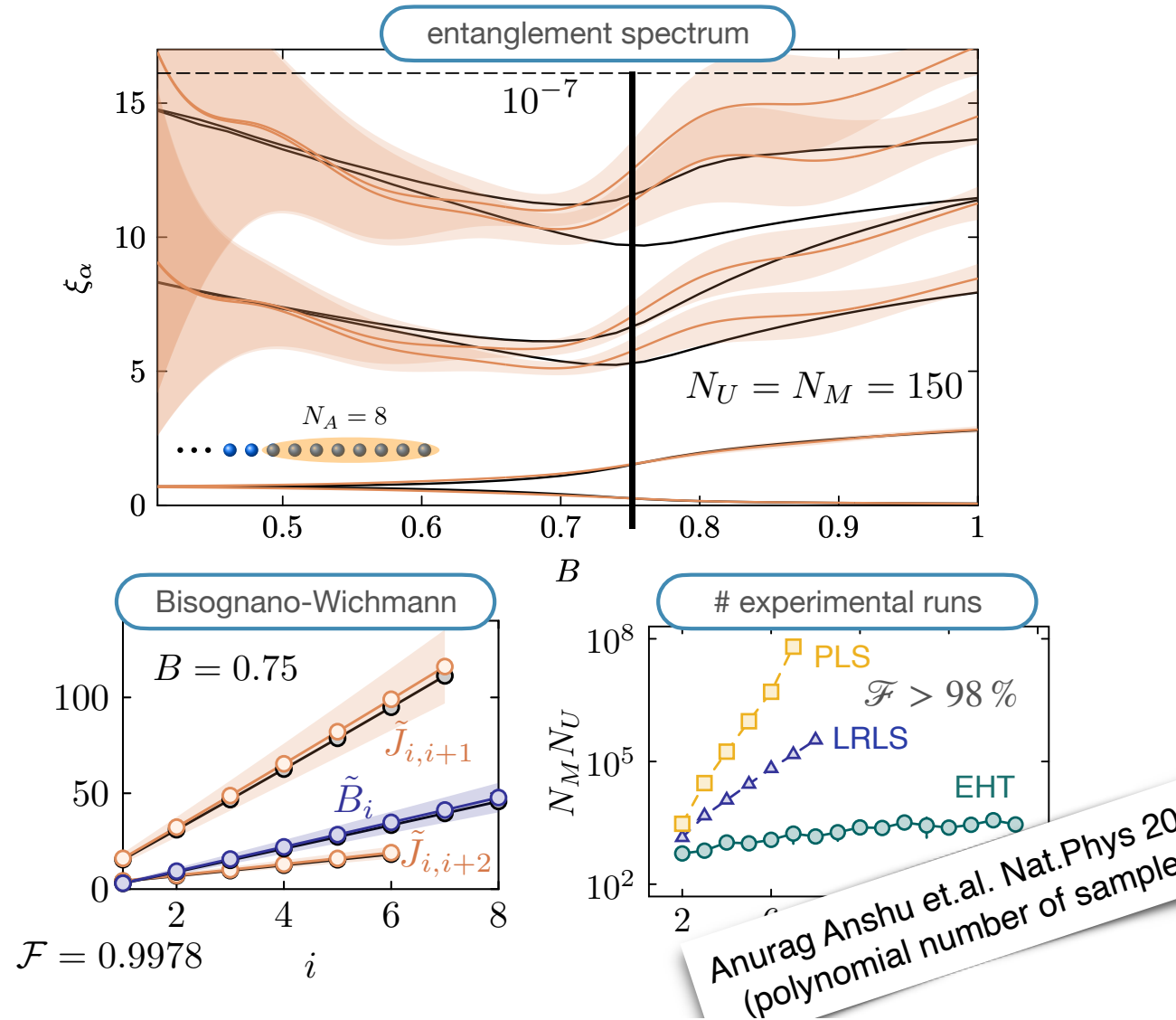
$$H = \sum_{i,j>i} \frac{J_0}{|i-j|^\alpha} \sigma_i^x \sigma_j^x + B \sum_i \sigma_i^z$$

$\alpha = 2.5$

Ansatz for Entanglement Hamiltonian

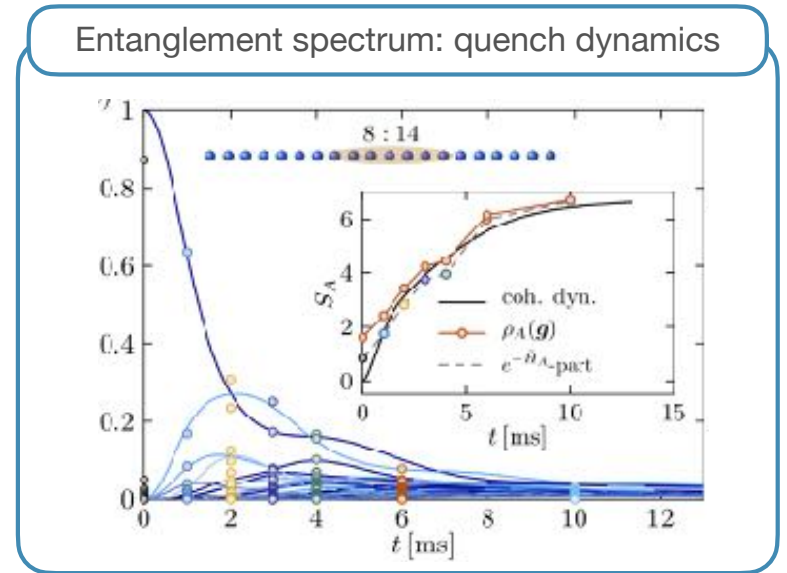
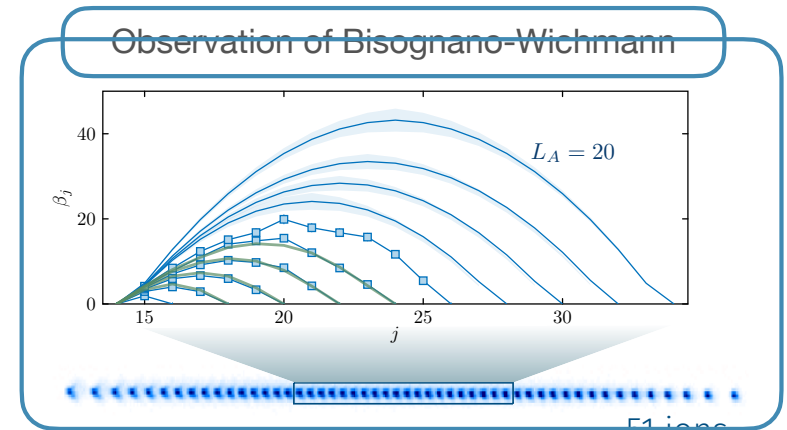
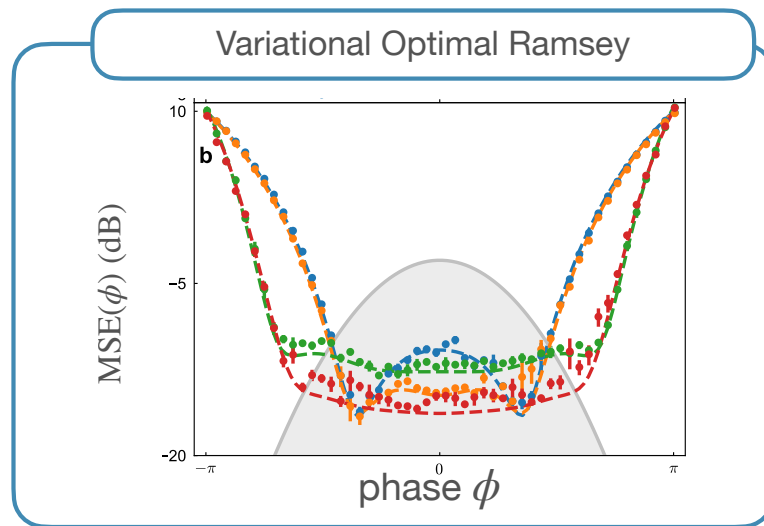
We choose a simple deformation of the system Hamiltonian

$$\tilde{H}_A = \sum_{i,j \in A} \tilde{J}_{ij} \sigma_i^x \sigma_j^x + \sum_{i \in A} \tilde{B}_i \sigma_i^z$$



Conclusions & Outlook

- Programmable Quantum Simulators with Atoms
- Hybrid Classical-Quantum / Variational Algorithms
- Randomized Measurements Toolbox
- Programmable Quantum Sensors



The Team

Theory:



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Experiment:



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F. Kranzl



C. Roos



R. Blatt